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RESEARCH ARTICLE

Smart Wheelchair Controlled Through a Vision-Based Autonomous System

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ABSTRACT People are preoccupied with their own duties in today's environment, making it tough to oversee and care for disabled people. Paralyzed and disabled people, on the other hand, have a powerful desire to move around freely. Different initiatives have been taken in the past to improve and preserve the self-esteem of such people, and various technologies have also been created to assist them in a better way. The primary advantage of a vision-based autonomous wheelchair is that it can provide greater independence to users with limited mobility. The methodology of other smart wheelchairs varies depending on the specific technology used. Some smart wheelchairs may use sensors to detect obstacles and provide feedback to the user, while others may incorporate GPS and other navigation technologies to assist with indoor and outdoor navigation. Our study proposes a revolutionary implementation of an autonomous system for fully handicapped people, allowing them to drive wheelchairs using just their eyes. The wheelchair's orientation will be determined by the orientation of the eye. The camera will follow the movement of the eyes. Furthermore, sensors will be installed on the front side of the wheelchair to identify any obstacles along its path. The calibration of the camera with the human eye, without obstructing its vision, is the most challenging task in this research, besides controlling the wheels of the wheelchair.

INDEX TERMS Image processing, smart wheelchair, robotics, eye motion tracking, Hough transform, Haar classifier.

I. INTRODUCTION

The rapid ageing of society in recent decades has contributed to growing demand for life support and healthcare. The ageing process in the elder people reduces their functionality and mobility, thereby affecting personal care and people's independence. The secret to freedom is their mobility. This is prone to physical and mental health changes which is one of the most crucial factors in assessing the person's functional capacity. We have seen in the media about manipulating an entity through the focus of one's eyes. Although much of the concept of eye control has been seen as humorous and fictional, it is useful not only for the future of natural inputs but also particularly for people with disabilities.

During a recent survey conducted by Center for Disease Control and Prevention (CDCs) in USA, they find out that one in four persons with sixty-one million people has disabilities, which have a major effect on their daily lives [1]. It means that people find it difficult not only to talk but also to travel. The only way for a person to get around here is rely on a person (nurse) to take care of the wheelchair during their movement. Another choice is to get a wheelchair which is mounted

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on a joystick [2], [3], thereby controlling the movement of the wheelchair. However, a joystick-based wheelchair is ineffective in case of complete paralysis [4]. Hence, the only option in these situations would be to use the support of a doctor to push the wheelchair around. Moreover, [5] disabled persons confront several challenges when it comes to getting work. They have inferiority complex within them that they cannot do any job in industry as nobody will hire an unfit man. Our idea of project will help them in many ways. Therefore, we developed a special form of wheelchair that will help these people to become self-reliant and incorporate them into our community [6].

Our project aims to use our eyes to monitor any entity that we see fit, in particular targeting the movement of the handicapped. The controller, yet we wanted a wheelchair; we could have interfaced with a motor vehicle [7], a fly, a device used for entertainment or a video game. The USB-exit from the laptop is taken from the microprocessor and the signal from it is used to control the wheelchair [8]. For the system to function properly, different sensors are connected to the Raspberry Pi [9]. Programming a Raspberry Pi to control a smart wheelchair involves several steps, including, but is not limited to, selection of the appropriate hardware components, setting up the Raspberry Pi, programming the microcontroller, and designing a controller. First, the hardware components required for the smart wheelchair include a motor controller, motors, batteries, sensors, and a Raspberry Pi. Once these components are obtained, the Raspberry Pi must be set up with an appropriate operating system, such as Raspbian, and any necessary software libraries and drivers.

Next, the microcontroller must be programmed to receive commands from the Raspberry Pi and translate them into motor movements. The controller must be designed to enable the user to control the wheelchair. There are several programs that can be used for controller design, including MATLAB, Simulink, and LabVIEW. The choice of program depends on factors such as the user's familiarity with the program, the complexity of the controller design, and the availability of necessary libraries and tools. Ultimately, the choice of program for controller design will depend on the specific requirements of the smart wheelchair project, as well as the user's expertise and resources.

We would like to admit that the most difficult part of the project was integrating the motors with the wheelchair, as each motor has two motor drivers, one on each side. This driver is made up of an H-bridge [10], which drives the engine based on the performance of the microprocessor. The drivers control the direction and speed of the wheelchair so that it can be moved forward, back, left, or right, accordingly. The embedded device [12], which is based on raspberry tubes, is used to transmit information to the sensor from the control system between the wheelchair and the tubing panel.

Having these facts into consideration, there are four options for fundamental movements for the user of a wheelchair: Forward, Backward, Right, and Left. Someone who does not have the expertise or mobility to operate a mobility scooter, for example, due to arm, hand, shoulder, or more general limiting diseases, and does not have the leg strength to boost a manual chair with their feet, can use a power chair [13]. Besides, a vision-controlled wheelchair presents a solution to all these kinds of problems. Therefore, we resort to these tasks, keeping into consideration the financial impacts and constraints of the application.

Most vision-based smart wheelchairs have the potential to enhance the accessibility of built environments for people with mobility impairments [19], [20], [21]. The theoretical implication is that individuals with disabilities can achieve greater independence, freedom, and participation in society. These wheelchairs raise questions about human-machine interaction, as users need to communicate with the device effectively. Theoretical implications include the study of user interfaces, control systems, and the adaptation of technology to suit individual user needs and preferences. The use of vision-based systems requires users to process visual information and make decisions, which can be cognitively demanding. Researchers can explore how to reduce cognitive load and optimize the design of these systems for user comfort and efficiency. The development of vision-based algorithms for navigation and obstacle avoidance is a theoretical challenge. Research in this area can lead to improved machine learning models, which may have broader implications for autonomous vehicles and robotics.

Many of the practical implications extend to how society perceives and accommodates individuals using vision-based smart wheelchairs [22], [23], [24]. These wheelchairs can provide users with a better understanding of their surroundings. They can identify the location of important landmarks, such as entrances and exits, and even provide information about the location of objects or people. The practical implication of this prototype is that it can make navigation in complex, crowded, or unfamiliar environments safer. It can help users avoid collisions with people and objects, reducing the risk of accidents. In conclusion, vision-based smart wheelchairs have significant theoretical and practical implications that go beyond simple mobility assistance [25], [26]. They have the potential to revolutionize the lives of people with disabilities, enhance research in artificial intelligence and robotics, and contribute to a more inclusive and accessible society.

II. LITERATURE REVIEW

A thorough review of the literature reveals that wheelchairs can be divided into two types; the one with a battery and another one with manual feeding [3], [14], [15], [16]. Although enough electric wheelchairs are available in the market, there is limited scientific published research on manual wheelchairs or the ones with batteries. The eye-controlled wheelchair [17] aims to help people with any disabilities, specifically regarding mobility. It can facilitate people who are suffering from locked-in syndrome, by making them independent in their movement.

Various types of wheelchairs have been designed to maintain a certain degree of independence of the disabled person [18], [20]. Controlled wheelchairs have been built for patients with foot, hand, voice recognition, head motion and eyeball movement [27], [28]. Different techniques have been utilized to propose a comprehensive solution for people who are suffering from various kinds of disabilities and paralysis. This paralysis might have been caused from accidents, stokes and head injuries which include the injuries on spinal cord. These paralyses are of diverse types i.e., Quadriplegia, Hemiplegia, Monoplegia, Diplegia and Paraplegia [29].

Therefore, we have studied different kind of wheelchairs before implementing our project [30], [31], [32]. A manual wheelchair is a freely moving device in which the patient sits. This vehicle [33] can be driven by hand by turning the wheels or by various programmed methods. Wheelchairs are used by those who find it difficult to go outside due to disease (either physiological or physical), accident. A single-arm drive [34] allows the user to move on a square line to the left or right, whereas a two-arm drive allows the user to move forward or backward. Another frequent type of wheelchair is a lever-drive wheelchair [35]. The user of this chair can go forward by pushing the lever back and forth. Lever-drive wheelchairs can be harder to manoeuvre in tight spaces due to their manoeuvrability issues.

In another model [36], the mobility of a wheelchair is controlled by a voice recognition system. This device is designed for those who have lost control of upper extremity vertebrate body parts like their arms due to an injury, sickness, or disability. Voice instructions can be used to communicate with this sort of system [37]. Overall, voice-controlled wheelchairs can offer a valuable alternative to traditional input methods, providing users with increased mobility and independence. However, users should consider the potential limitations and drawbacks before deciding if a voice-controlled wheelchair is the right choice for them. For instance, External noise or other voices could potentially interfere with the operation of a voice-controlled wheelchair, causing it to respond incorrectly or not at all.

One scheme suggests that the wheelchair can be controlled by head movement. This model is useful for those people who cannot move the wheelchair with the help of their hands due to the lack of power in arms or psychomotor problems [38]. In this model, wheelchair can be operated with the help of accelerometer. This accelerometer can control the movement of a wheelchair according to the head movement [39]. Similarly, a wheelchair can move by following the lines that are drawn on the ground. In this system color sensor or infrared sensor is being used to follow the path. Color or infrared sensor just detects the color of line and moves the wheelchair accordingly [40]. Practically this method cannot be implemented on wheelchairs as it requires the lines on ground to follow.

This mechanism allows the wheelchair to be controlled with three fingers. The wheelchair's speed is controlled by one finger, while its orientation is controlled by other two fingers. There is a lot of room for error; we can only use two fingers instead of three to manage the speed, and the system does the rest [41]. The control of an Electric Powered Wheelchair (EPW) using Electromyography (EMG) is classified into two categories: pattern recognition and pattern less recognition. A classification is employed in EMG systems based on pattern recognition to identify or classify the signal patterns taken from raw dataset as functions [42], [43].

Simultaneous localization and mapping (SLAM) [44] model is used in navigation, robotic mapping and for virtual mapping. Simultaneous location and mapping are properties that solve the difficulty of creating or updating a map of an unknown area. In this model, an object moves along the unknown path and according to the property of SLAM algorithm, the object generates the map of that location by moving from one point to another [45].

Another system is proposed in which a wheelchair is equipped with a sunshade, a foot pad, a head panel, and obstacle avoidance, all of which are independent of a person's ability to participate in society [46]. When the humidity sensor senses rain, heat, or cold, the head mat activates immediately. Furthermore, the ultrasonic sensor is used to identify obstructions using GPS location tracking. The proposed environment-based system interacts successfully with a living environment, as demonstrated by a prototype system implementation [47].

Similarly, a wheelchair for quadriplegia patients [48] utilizes an eye blink sensor, a joystick and keypad module. Quadriplegia is the paralysis of the trunk, legs, and arms from the neck down. The condition is usually caused by an injury to the spinal cord, which contains the nerves that transmit movement and sensation messages from the brain to various parts of the body. Furthermore, the disabled person is given more independence by using additional sensors such as a heartbeat sensor and a temperature sensor, which constantly monitor the patient's health situation [49]. A urine level indicator is also used to keep the patient comfortable. If a patient falls while using a wheelchair, the wheelchair's fall detection system recognizes it. During an emergency, all information may be conveyed to medical workers and the patient's guardian so that they may respond quickly [50].

III. METHODOLOGY

Prior to the implementation of our project, we went through the process of researching several types of wheelchairs ([3], [4], [5], [6], [7], [8], [14], [17], [18], [20], [40], [41], [56]). A wheelchair that is manually operated is a device that allows the user to sit in it and move around freely. This vehicle can be driven manually by spinning the wheels, or it can be operated by a variety of different techniques that have been programmed. As a result of a disability, an accident, or a sickness (either physiological or physical), wheelchairs

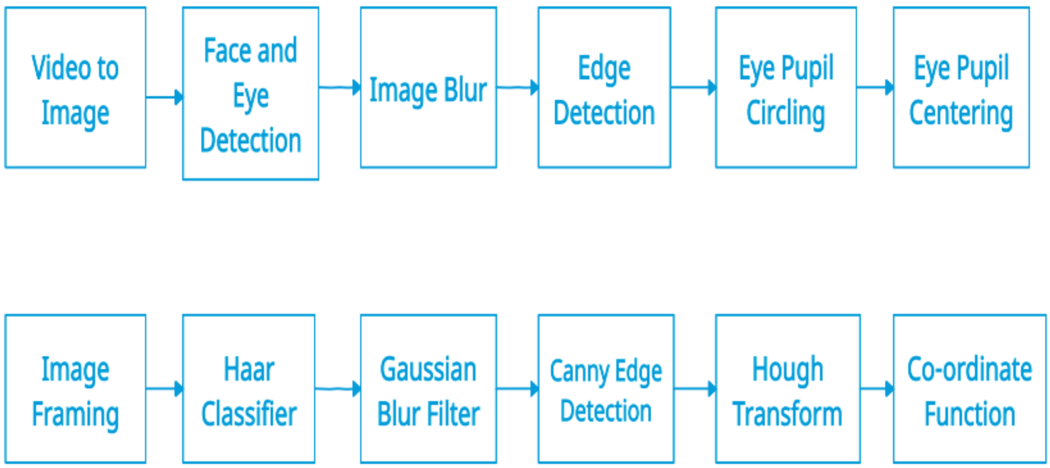

FIGURE 1. Workflow diagram for detection of Eye pupil direction through Image processing.

are utilized by individuals who have difficulty going outside. Additionally, the wheelchair can be controlled by the user's head movement in a different model. Due to a lack of power in their arms or psychomotor issues, individuals who are unable to move their wheelchair with the assistance of their hands can benefit from this model.

This project is divided into three parts. The first component is the camera, which is positioned in front and uses a variety of image processing techniques to record images of either of the eyes (left or right) and monitor the location of the eye pupils. The second one constitutes wheelchair motors. These motors are instructed to move left, right, or forward based on the location of the eye being controlled by the motor drivers. Last, the Raspberry-Pi board, which is used to do image processing and control the hardware system, is the most vital component of the hardware system. It records visual frames in real time and creates commands based on eye movements [51]. Based on these commands from the processor, it will start and stop the motors and either move them in clockwise or counterclockwise direction.

In order to give critical input to the wheelchair's system to function properly, pressure and obstacle detection sensors have been attached to the microprocessor in addition. The final phase in the design process is to connect the wheelchair to the motor drivers. Each of the two motors on the left and right sides will have two additional motors. Each motor driver will feature an H-bridge that will power up the motor based on the microprocessor's output [52]. Motor drivers will assist in controlling the pace and direction of the wheelchair's motors, allowing it to travel forward, backward, left, or right.

The Raspberry Pi will be used as the wheelchair controller in our hardware. It collects data from the laptop, ultrasonic sensors, and motors, among other sources. It will produce the proper actions based on those inputs, such as waking up, moving ahead, right, or left, and maintaining a consistent pace. For our hardware goal, instead of using the wheelchair's on-board controls and drivers, we'll design and utilize motor drivers to manage the wheelchair's speed and direction. The motor driver will run the wheelchair using a Pulse Width Modulation (PWM) signal from the microcontroller [53]. Our software objective will also be to use Open CV on our laptop to analyse the images captured by the eye-tracking camera. We developed a tracking algorithm that allows us to determine the eye's position as well as whether it is open or closed. After it has been determined, it will send signals to the microcontroller.

The main objective of microcontroller is to process the raw data and take decision from the outcomes based on the user response. For instance, as soon as the user changes its vision, the wheelchair should also move its direction with minimum amount of response time. Figure 1 demonstrates the workflow of our system from taking an input signal from a video and detecting the direction of wheelchair based on pupils' movement.

The system's operation initiates with the regular capture of photos from a webcam, forming the foundation for its functionalities. To enable facial recognition, the Haar cascade approach is ingeniously employed, seeking to detect faces within the images and subsequently localizing the eyes within the specified facial regions. Similar to its face detection counterpart, the Haar cascade technique is further utilized to identify the eyes, which are subsequently encompassed by rectangular boxes for easy identification. Now, the system shifts its focus to identifying and characterizing the crucial areas of the eye pupils. An array of image processing techniques, such as blurring, color conversion, thresholding, filtering, edge detection, and the powerful Hough transform, are seamlessly integrated into the system to achieve this task. Notably, the Hough transform technique proves instrumental in detecting circles, allowing for accurate identification of the eye pupils [54]. Circles are detected using the Hough transform technique. The pictures on the Raspberry Pi are taken with a USB webcam. All OpenCV libraries for image processing are placed on the Raspberry Pi RAM. It will process and work in that region without any processing delays.

The visual data for analysis is captured through a USB webcam on the Raspberry Pi, ensuring a practical and convenient setup. All the necessary OpenCV libraries for image processing are efficiently allocated within the Raspberry Pi's RAM, ensuring swift and uninterrupted processing of the visual data. The system showcases remarkable efficiency, swiftly and meticulously analyzing each frame without any noticeable processing delays, thereby guaranteeing real-time responsiveness.

Simultaneously, the system remains vigilant and proactive in its surroundings. It diligently scans the environment, constantly on the lookout for obstacles lying ahead, with the primary goal of keeping the pathway clear and safe for the wheelchair's seamless movement. By continuously analyzing the camera feed in real-time, the vision system diligently detects and identifies various objects that come into view. Upon identifying an object, it promptly classifies it into distinct categories, such as pedestrians or obstacles, accurately gauging its potential impact on the wheelchair's navigation in the dynamically changing environment.

Through the harmonious integration of advanced image processing techniques and robust object detection capabilities, this vision-based smart wheelchair system demonstrates its prowess in ensuring not only efficient eye detection for facial recognition but also proactive obstacle avoidance and clear pathway maintenance. Its ability to respond promptly to dynamic changes in the environment makes it a reliable and secure companion for users, providing them with a sense of independence and confidence in their daily mobility.

The flowchart of the system's execution in figure 2 is depicted in the figure 3. First of all, the system will recognize a face and an eye in the image with the help of Haar features classification. If the Eye is open and no obstacle is being detected by the system, the wheelchair will follow the pupils' direction of the user. If the user looks forward, the wheelchair will also move forward. similarly, If the user turns his head to the left or right, the wheelchair will do the same motions and will start moving in that particular direction accordingly.

The system began with a camera which is continually recording photos, processed on the Raspbian operating system. A USB camera was used to capture the image at high pixel rates. If the eye does not move, it is deemed open. As soon as the power is turned on, the system will begin to operate, and it will operate according to the command settings. This system is built on a real-time data capture operating system. The camera module is attached to the user's eye. In addition, the distance between the eye and the camera is controlled. It might be anything from ten to fourteen centimeters in length. The camera will take photographs of the user's face and figure out where the pupil is located. After recognizing pupil position, the system algorithm calculates the average value in the center of the eye. This provides accurate information about eye movements. We shall use VNC (Virtual Network Computing) Viewer in order to see the images taken by camera [55]. Following steps should be taken in order to interface it. The first step is to install an operating system on the micro-SD card (memory device). Using Win32 disk-imager software, a Raspbian image file was booted. When a Raspberry Pi board is attached to a bootable memory device, the operating system may be accessed without needing to reboot [56].

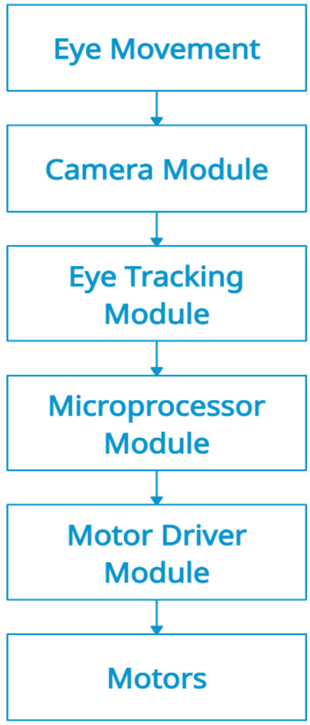

FIGURE 2. Functional block diagram of the proposed system.

After capturing the picture, it is crucial to eliminate noise and haze caused by environmental factors affecting the camera sensor. Several steps are involved before reaching the classification stage. The image is imported as arrays and standardized to a specific size. Gaussian Blur is applied for noise removal, smoothing the images and reducing their size. This low-pass filter eliminates high-frequency signals. Next, image segmentation separates the background from objects, enhancing the region of interest, resulting in a better pre-processed dataset for further categorization through the classification process.

We use the free OpenCV library algorithm for image processing. The OpenCV library is particularly useful since it has image processing capabilities. A one-of-a-kind system execution algorithm was used to conduct the technique. To establish the pupil center point of the eye, we used a variety of techniques:

A. FACE AND EYE DETECTION

Face and eye detection may be done easily with the OpenCV library. The first Haar cascade approach is used to identify both faces and eyes. The user's face is detected by a system camera. After recognizing the eye, the system used the Haar

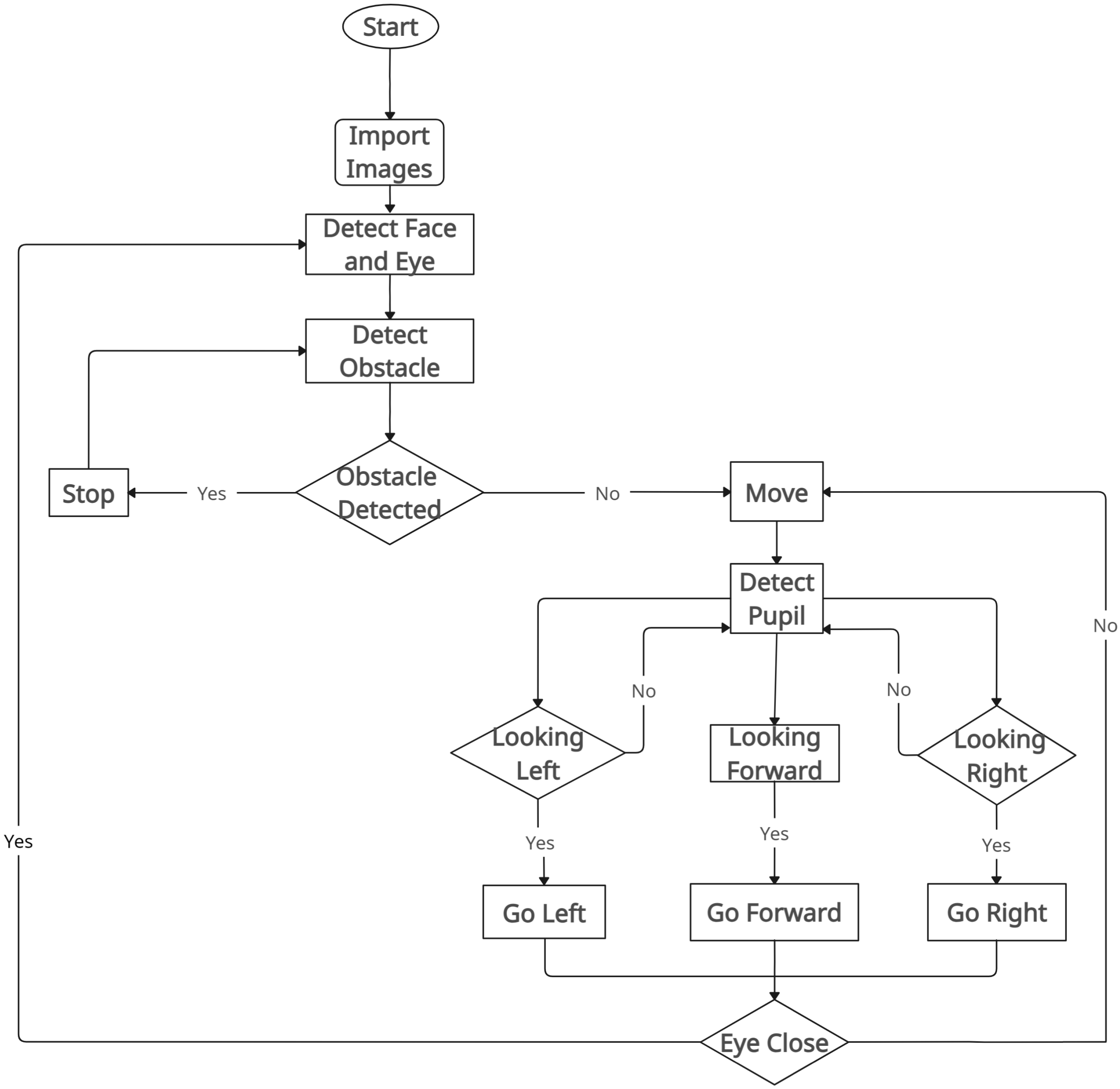

FIGURE 3. Flowchart of the working system.

cascade technique to discover and label the ocular area. Furthermore, the technology accurately identifies both eyes based on their distances.

B. IMAGE BLURRING AND FEATURES DETECTION

The image is blurred using the Gaussian blur filter. This aids in detecting the exact margins of a cropped image's specified region.

C. EDGE DETECTION AND EYE PUPIL CIRCLING

As an outcome, the system uses canny edge detection and corner edge identification algorithms to estimate the boundaries of the eye pupils. Following that, a circle is created on the eye pupils using the Hough circle transform arithmetic function.

D. HOUGH TRANSFORM

We apply the Hough circle transform technique to construct a circle on the eye pupil over the edge detection resulted images. The Hough transform identifies the motions of the eye pupil and creates circles based on continuously collected pictures from the camera.

E. EYE MOTION TRACKING

A coordinate system is used to track eye movements. The horizontal axis, represented by X, and the vertical axis, denoted by Y, represent left and right eye motions, respectively. As a result, the coordinate points (X1, Y1) are used to determine the position of the eye pupil in the following manner:

$$X1 = ((X0 + X2))/2 \tag{1}$$
$$Y1 = ((Y0 + Y2))/2 \tag{2}$$

The method displays the basic eye pupil motion detection software, which pushes the wheelchair in left, right, and forward directions, depending on where the eye pupil is located.

IV. HARDWARE IMPLEMENTATION

The system is meant to be completely autonomous with all modules operating independently of one another. The microcontroller is coupled to a camera and a Wi-Fi module. In our project, we utilize a Raspberry Pi 3b+ to gain faster processing and reduced lag in the system (figure 4).

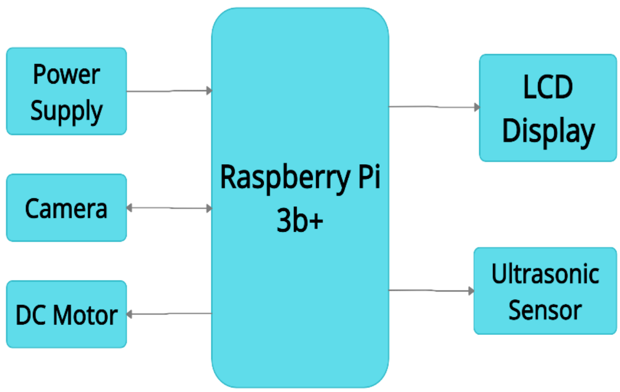

FIGURE 4. Block diagram of the proposed system.

A. HARDWARE COMPOSITION

Any electronic system requires power as a basic necessity. Individual components must be given the appropriate power supply in this system. The Raspberry Pi, camera, sensor, and motors should all be powered by the standard power supply. Using an LCD screen on a wheelchair can provide several advantages, even if it might seem unnecessary at first glance. Though it is true that some people with disabilities may have difficulty moving their heads, yet the benefits of an LCD screen can still be helpful for many wheelchair users. An LCD screen can be used as an augmentative and alternative communication device for people who have difficulty in speaking. With the help of a touch input mechanism or other assistive technologies, users can convey their needs, emotions, or preferences to others. The complete functionality of the implemented system is represented in figure 5.

Power is an essential component of any electronic system. In order for this system to function properly, each component possesses its own unique power supply. The standard power supply ought to be used to supply power to the Raspberry Pi, the camera, the sensor, and the motors. Even though it would appear to be unneeded at first glance, using an LCD screen on a wheelchair can provide one with a number of benefits. The advantages of an LCD panel might still be helpful for a great number of wheelchair users, despite the fact that it is true that some people with impairments may have trouble turning their heads. Individuals, who have trouble in speaking, may benefit from the usage of an LCD screen as a non-traditional and supplementary communication tool. Individuals are able to communicate their requirements, feelings, or preferences to other people by utilizing a touch input mechanism or other assistive technology. Figure 5 shows a representation of the system's full capabilities after it has been developed.

A Wi-Fi module is available to us which enables the Raspberry Pi to establish a connection to the internet and communicate with other devices in a variety of crucial situations. It allows the Raspberry-Pi to connect to the internet and interact in several important circumstances. Individual parts, such as the Raspberry Pi, camera, sensor, and engines, can be powered using the power circuit. Because of the quality of our project design, any conventional wheelchair on the market may be converted into an electric wheelchair. We went shopping and purchased a standard light weight wheelchair. Two steel plates were welded to the rear of the wheelchair so that motors could be mounted to them. Chain and Sprockets are used to move the wheelchair from one point to another point. The chain is moved along the sprocket and the rotating shaft of Wiper motor. One end of chain is on sprocket of wheelchair and the other end is on the rotating shaft of motor. Rotating shaft converts the electrical energy of motor in the linear or mechanical energy.

Similarly, wiper motors are being used for producing linear force to move the wheelchair. A wiper motor is an electrical device that operates on power supply in order to move the wiper blades of motors which are inserted in motors and work in smooth motion by converting electrical energy into mechanical energy. An electric generator, on the other hand, converts mechanical energy into electrical energy in the opposite way. Electric motors generate torque as a linear or rotating force. Like other motors, a wiper motor can also move in one direction, and can be converted into back and forth movement as per requirements. Two Wiper motors are therefore used, one for each wheel.

Ultrasonic sensors are placed around the wheelchair to provide comprehensive obstacle detection and collision avoidance. We have mounted the sensors on the front of the wheelchair that will help in detecting obstacles which are directly in the path of the wheelchair, allowing the user or wheelchair's control system to make adjustments to avoid collisions. This will also allow the system to detect any obstacle or hurdles in the pathway. Similarly, we have mounted a sensor on the back of the wheelchair which will provide coverage for obstacles behind the wheelchair when moving in reverse or making turns. This will ensure that the wheelchair can safely navigate through different

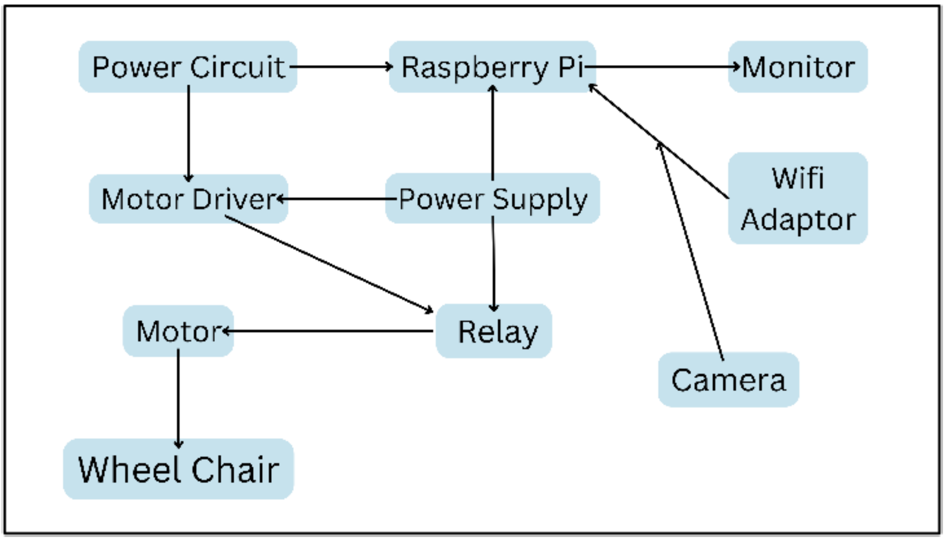

FIGURE 5. **Design model of the proposed system.**

environments and detect obstacles in real-time, reducing the risk of accidents and providing a more comfortable experience for the user.

B. SOFTWARE COMPOSITION

The system is built on the basis of real-time data set. Following is the functional block diagram of our project in figure 2 that describes the workflow of the project. This was thoroughly followed for designing of the project.

Moreover, we have used Raspberry Pi 3B+ module as a microcontroller in our system which is the most recent version in the Raspberry Pi 3 series, with a 64-bit quad-core CPU operating at 1.4GHz. A BTS 7960 is integrated with the system to control motor's speed and rotation. BTS 7960 is half bridge motor driver which consists of four pin that are used for power supply and other four pins that are used for connecting as input and output source with the Arduino. In case of half bridge driver, only one motor can be controlled because it has two pins as power supply to motors with positive and negative terminals. So, in that case two motors will be controlled with the help of two drivers. However, just one BTS 7960 driver is utilized to operate the two motors in that project.

When a high-voltage electrical pulse is supplied to an ultrasonic transducer, it vibrates in a certain frequency range, resulting in an explosion of sound waves. When an obstacle is in front of the ultrasonic sensor, the sound waves are reflected as an echo, resulting in an electric pulse. This mechanism determines the time elapsed between delivering sound waves and getting the echo. To establish the state of the detected signal, the echo models are compared to the sound wave models.

In the proposed system, vision-based smart wheelchair is equipped with cameras and smart vision algorithms to process visual information from its surroundings. When an object suddenly comes into view, the system would typically follow these steps to react appropriately.

1) *Object Detection*
   The vision system continuously analyzes the camera feed to detect and identify objects in its path. The object detection algorithm that we used comprises of YOLO (You Only Look Once) algorithm to detect the obstacles ahead and Haar features to detect the user on wheelchair.
2) *Object Classification*
   Once the system identifies the object, it classifies it into different categories (e.g., pedestrians, obstacles etc.) to understand its potential impact on the wheelchair's movement in real time environment.

Whenever the system detects a potential collision, it engages the collision avoidance module to stop the wheelchair instantly until the next movement command is given by the user. Moreover, in case of an object suddenly coming in the pathway of wheelchair, multiple Ultrasonic sensors are being used in the system, which will help the system to detect the upcoming obstacle immediately and stop the wheelchair from collision.

To complete this procedure, a camera module is positioned in front of the eyes (left or right) to record pictures and monitor the location of eye pupils using a variety of image processing algorithms, as shown in figure 2. The wheelchair motors are instructed to go left, right, or forward, depending on where the eye is located. The hardware system's most vital component is the Raspberry-Pi board, which processes pictures and controls the hardware system. It records visual frames in real time and creates commands based on eye movements. The Raspberry Pi then sends control signals to the engines, telling them to perform things like turn the motors clockwise, anti-clockwise, or stop.

C. DESIGN CONSIDERATIONS

Vision-based smart wheelchairs offer users an improved awareness of their surroundings ([5], [7], [9], [11], [20]). They can recognize important places like entrances, exits, and even provide details about where objects or people are located. The practical significance of this innovation lies in its capacity to enhance safety when navigating intricate, crowded, or unfamiliar settings. By helping users avoid collisions with people and obstacles, these wheelchairs reduce the risk of accidents. To sum it up, vision-based smart wheelchairs have both theoretical and practical implications that extend beyond just assisting with mobility. They hold the potential to transform the lives of people with disabilities, advance research in artificial intelligence and robotics, and promote a more inclusive and accessible society.

Vision-based smart wheelchairs utilize cameras and sensors to perceive and analyze their environment [37], [40], [41], [42]. They face limitations due to unfavourable visibility conditions, such as fog, intense rainfall, or reduced illumination, which can provide challenges, and in certain situations, even render navigation impracticable. Manoeuvring in interior spaces, particularly in congested or intricate environments, can provide challenges. Vision-based systems may encounter difficulties in detecting little impediments on the floor, such as toys or wires, which could pose a risk to the user. Wheelchair-mounted cameras have a limited field of vision and may fail to detect obstacles or objects that are beyond their reach. When making a turn or altering one's course, there is a possibility of colliding with objects or individuals, perhaps leading to accidents. Due to the intricate technologies employed, the proposed system may incur significant expenses.

Cameras and sensors are used by vision-based smart wheelchairs to "see" their surroundings ([19], [22], [23], [24]). They are constrained by bad visibility circumstances, such as fog, heavy rain, or low light, which can make navigation difficult, if not impossible, in some scenarios. It can be difficult to navigate around interior areas, especially in congested or complicated settings. Vision-based systems may struggle to identify minor obstructions on the floor, such as toys or wires, which might endanger the user. Cameras on wheelchairs have a restricted field of view and may miss obstacles or items that are out of their range. When turning or changing direction, this might result in accidents with objects or people. Because of the complex technology used, the suggested system may be costly. This expense may make them inaccessible to certain people who may benefit from them. Finally, there is no standards for vision-based smart wheelchairs as of yet, which means that various models may have varying capabilities and limits. If a user switches to a different wheelchair, they may need to adjust to new systems. To summarize, while vision-based smart wheelchairs represent exciting technical advances, they are not without limits. These constraints frequently centre around their capacity to effectively see and respond to their surroundings, as well as privacy, usability, and cost issues. Addressing these constraints is critical to making this technology more accessible and useful for those who have mobility issues.

V. RESULTS AND DISCUSSION

The purpose of this project is to develop hardware and software components for a smart wheelchair that controls motion using eye movement. The system collects image processing result data based on the Eye pupil's center value signal and sends it to the motor driving circuit for wheelchair movement. This part mostly shows the prototype and experimental results. To begin with, the Raspberry-Pi controller may efficiently transmit signals to the motor driver circuit by analyzing the motions of the eye pupils, leading the wheelchair to travel in the right directions.

The OpenCV library is utilized directly for face and eye detection. Initially, the system is trained to capture the face from the scene through Haar features which has been applied in a similar approach as in Viola-Jones algorithm. The algorithm is trained on a lot of images with positive images which contain faces and negative images (which do not contain faces). The algorithm uses "detectMultiScale()" function in cv2 which returns the boundary rectangles for both detected eyes. A first Haar cascade approach is employed for both Face and Eye recognition. A system camera recognizes the user's face. When it detects the face, the system locates the eye and labels the ocular area using the Haar cascade approach. As shown in figure 6, the system identifies both eyeballs precisely based on their proper distance from each other.

Following the face and eye identification, the next step is to find the location of the pupil. For this, the system blurs the picture with a Gaussian blur filter to get the chosen region of interest in order to determine pupil movements and direction. This aids in detecting the user's gaze movement from the exact boundaries of a specified section of the cropped image.

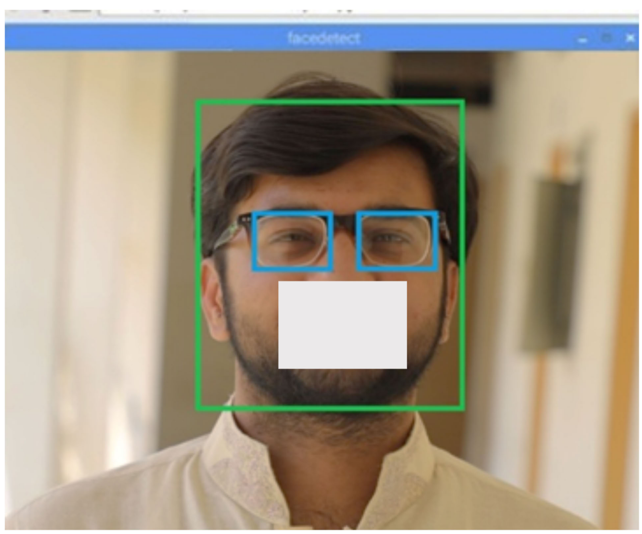

FIGURE 6. Face and Eye detection using Haar Classification.

Finally, Haar features have been used to detect both eyes in the picture. Furthermore, the circular Hough transform has been implemented to detect the pupils from both eyes in a frame. Once the system is able to detect the pupils, it can track the motion of pupils through which the direction of their movement can be identified using pupil tracking algorithm. This algorithm measures the coordinates of both pupils in each frame of image and detects the motion based on the difference between coordinates of both pupils. Figure 7 depicts this cropped vision of the eyes and pupil detection.

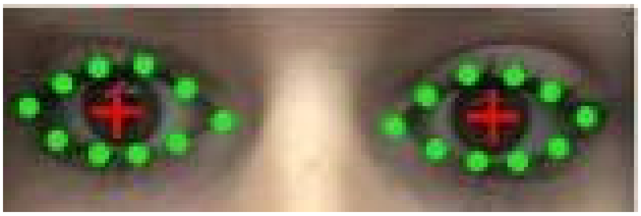

FIGURE 7. **Cropped vision of eyes and pupil detection.**

Similarly, figure 8 shows that the pupil's direction is being detected by the system. As the user is looking at right side, the system will send an immediate response to microcontroller to move the wheelchair in right direction. In order to improve the direction of eye pupil, we have used the dark-pupil recognition technique. Here, the algorithm detects the pupils using high contrast regions in the frame as the overall region around the eyes that has more light than pupils. Once the pupil gets detected, the algorithm starts tracking its motion by using a threshold difference in the images. Furthermore, circular Hough transform has been implemented to detect the direction of motion of pupil. As seen in figure 8, the algorithm initially detected the eyes through Haar features and clipped the position of pupils in the frame. After identifying the position of pupils, the system locks the coordinates of both pupils. When a person looks on either right or left side, the coordinates of both pupils change and hence the system detects the motion of pupils, keeping the coordinates of both pupils. Thus, these coordinates are further used to identify either a person is looking at the right direction or left.

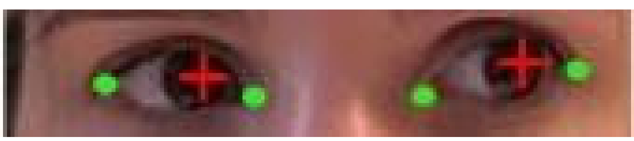

FIGURE 8. **Pupil's direction being detected by the system.**

Further results show the final working of the system using Eye Motion Tracking System. Here, the system will first identify the pupil's orientation and transmit a signal to the system's central processing unit. This will cause the wheelchair to turn in that direction. If the user looks in front of him, the wheelchair will move forward, whereas the wheelchair will move right and left in case the user looks in any direction. Figure 9 shows the pupil detection through Eye Motion Tracking System, and the corresponding result is being displayed on the screen as "looking left". In figure 9, the system initially detected the eyes of the user using Haar transform and a "plus sign" can be seen in both eyes which represents the pupil detection. At the top left corner, the system shows the current coordinates of both pupils in the form of Cartesian coordinates (x,y): the left pupil at (131,227) the right pupil at (217,223).

FIGURE 9. **Vision detection from the face through the system.**

Moreover, figure 10 shows that the system detects that as the user is looking in center, so the wheelchair will move ahead in forward direction. Along with the direction of

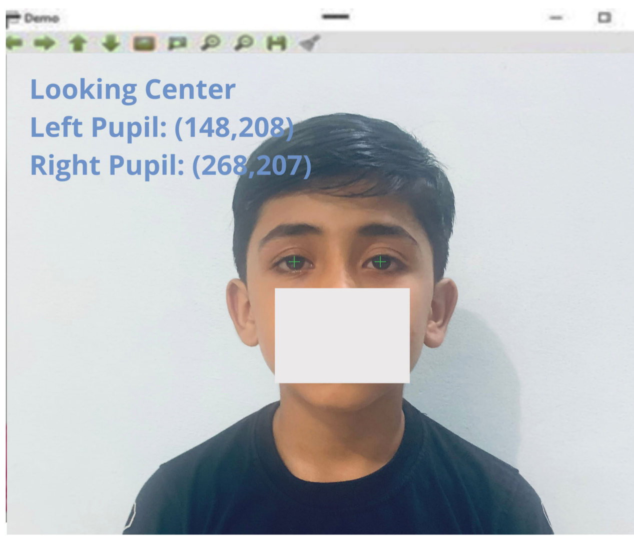

FIGURE 10. **System shows that user is looking in center so the wheelchair will move in forward direction.**

motion, the system also shows the coordinates of both eye's pupils from the origin, which is a very useful figure of merit.

Similarly, figure 11 shows that the system detects that as the user is looking in right direction, so the wheelchair will move in the right direction as well. Along with the direction of motion, the system shows the coordinates of both eye's pupils from the origin, like previous case. This is again a successful detection of the pupil of the eye, indicating the movement of the wheelchair in a direction that is desired by the user.

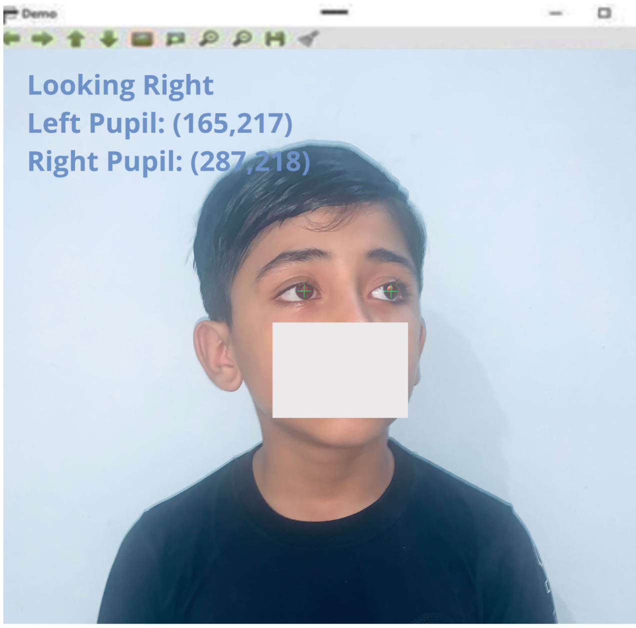

FIGURE 11. System shows that user is looking in right direction so the wheelchair will move in right direction accordingly.

We have designed our wheelchair that uses two high power dc geared motors. Raspberry pi was selected as microcontroller. Ultrasonic sensors are used to detect an obstacle. The model of the present wheelchair which is presented here, is sufficient and competent. It is capable of following any curve or cycle. Because points are granted depending on the distance travelled and the overall speed of the robot, we must create a lightweight and fast wheelchair. After discussions with the physicians in the orthopedic departments of two nearby hospitals, we employed two high-speed motors and a high-sensitivity sensor circuit.

Numerous factors had to be considered for this design. For instance, speed of the wheelchair is also affected by body weight and wheel radius. The last design phase is to connect motor drivers to the wheelchair. Each motor on the left and right sides will have two motors. Each motor driver will include an H-bridge that will power up the motor in line with the CPU output. Motor drivers will assist in controlling the pace and direction of motors, allowing the wheelchair to travel forward, backward, left, or right. Figure 12 shows a prototype implementation of the hardware design.

Figure 13 shows the proteus simulation for the proposed system. As it can be seen that two motors including left motor and right motor are connected to the H-bridge, which is further being controlled by microcontroller AVR. Two virtual terminals have been attached in order to check the output response of the system constantly.

After getting results from the prototype, we compared the accuracy of our system with an Eye controlled powered wheelchair and investigated the outcomes. It can be seen the comparison between proposed methodology and Eye controlled powered wheelchair in the figure 14.

During movement of wheelchair in forward direction, our proposed system got 91.3% of accuracy in comparison with the powered wheelchair having an accuracy of 91.5%. Furthermore, while moving in right direction, the powered wheelchair has an accuracy of 89.1% whereas our proposed system got 93.1% of accuracy which is a commendable improvement in results. Similarly, during taking turn into left direction, our prototype got an accuracy of around 93.7% whereas the eye controlled powered wheelchair has an accuracy of 86%. So, in conclusion to this comparison, our proposed system has better edge detection while taking turn in either directions left or right. And in forward movement there is a minimal difference in accuracy of both systems which is almost negligible in this case as our threshold speed is 5.5 mph to get better recognitions and smooth detection.

Similarly, figure 15 shows the recorded values on different intervals as the prototype has been tested on a designated pathway with little bumps on it. In first round, we tested the movement of our system in a straight direction and recorded its speed and multiple intervals. As it can be seen that initially the acceleration of wheelchair is increasing gradually and after reaching at the threshold of 5.5 mph of speed, it maintains its velocity. A little fluctuation can be seen throughout the whole track because of bumps on the road, a fact that cannot be avoided at this stage of development. Table 1 shows the movement and direction to correlate the eye pupils with the direction of movement in the mentioned cases, respectively.

To check the implementation, in second round, we have tested our system on a designated path with left and right turns where the user turns wheelchair in either direction by movement of his vision. In this type of movement, the differential system used between tyres helped the motors to balance powers in both tyres and give an adequate power distribution while turning into either direction. The differential system is working as backbone in the movement of smart wheelchair to move on bumpy and rigorous paths where tyres can actually underperform in a situation (an instance when one tyre is on high ground and other one is on low ground while taking turns). So, these results conclude the performance of our proposed system with astonishing efficiency.

However, if we talk about the precision and efficiency of the system, we tested the proposed system with a real time dataset of 111 pictures which were extracted through real time video while navigating the wheelchair. Out of 111 frames, the proposed system detected 103 frames accurately, which gives

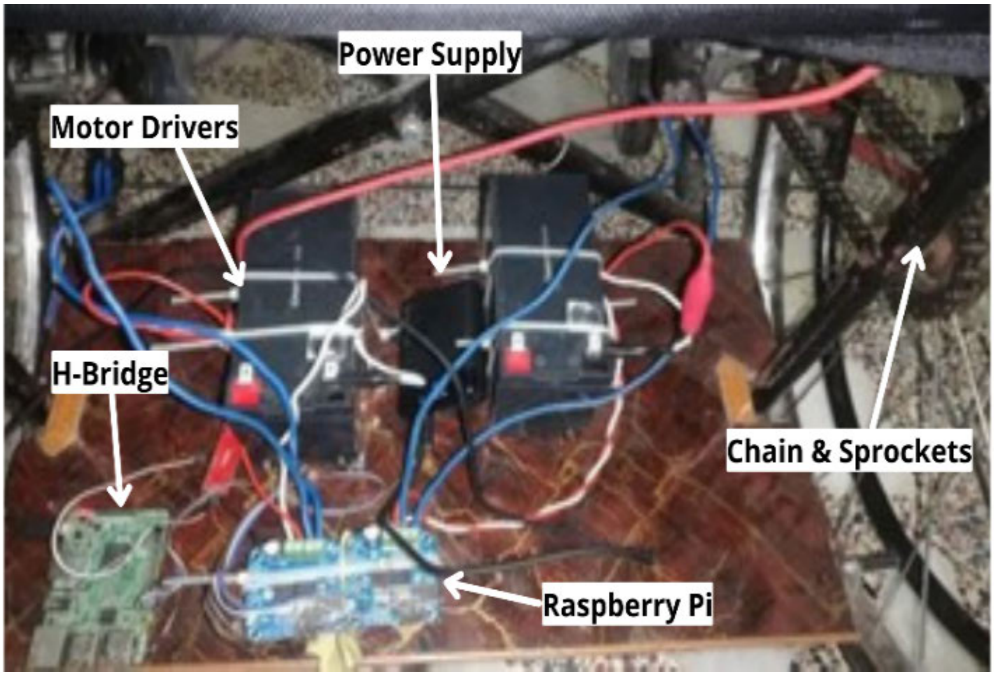

FIGURE 12. Hardware implementation of the prototype.

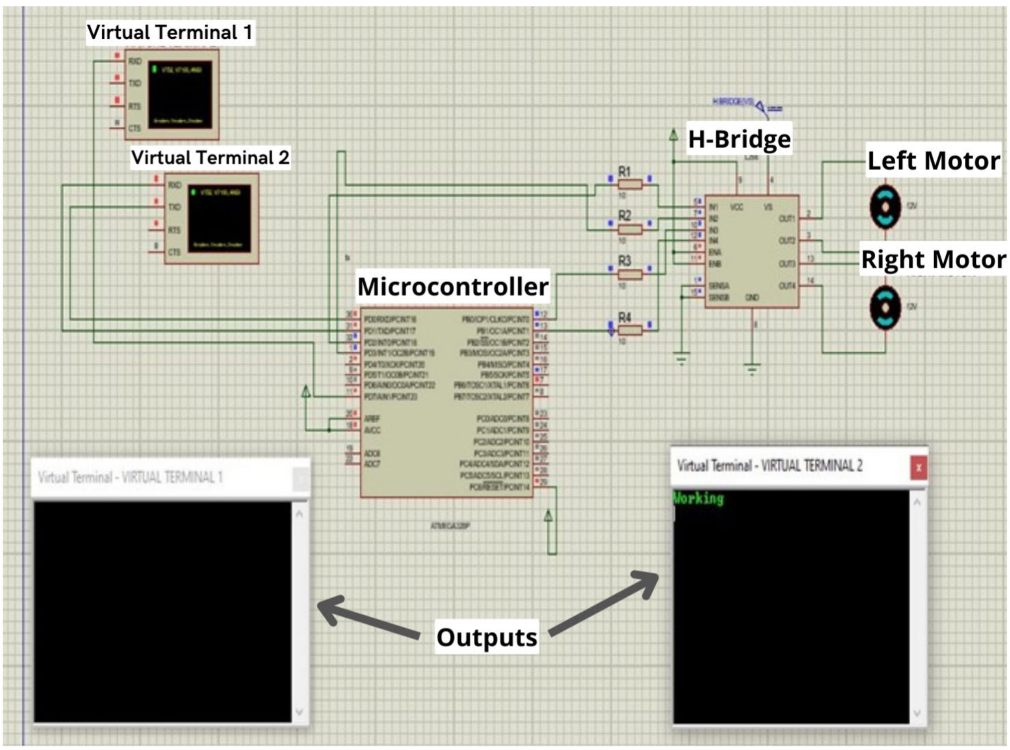

FIGURE 13. Proteus Simulation of the proposed system.

an accuracy of 92.8 percent and precision of 92.3 percent. Following is the confusion matrix obtained after testing out the dataset which illustrates the classification of different directions through pupil's movement as shown in figure 16.

A. COMPARISON WITH OTHER WORKS

We have implemented deep learning methodology to train the system. Deep Learning models have the capability to adapt and generalize well to different scenarios and conditions.

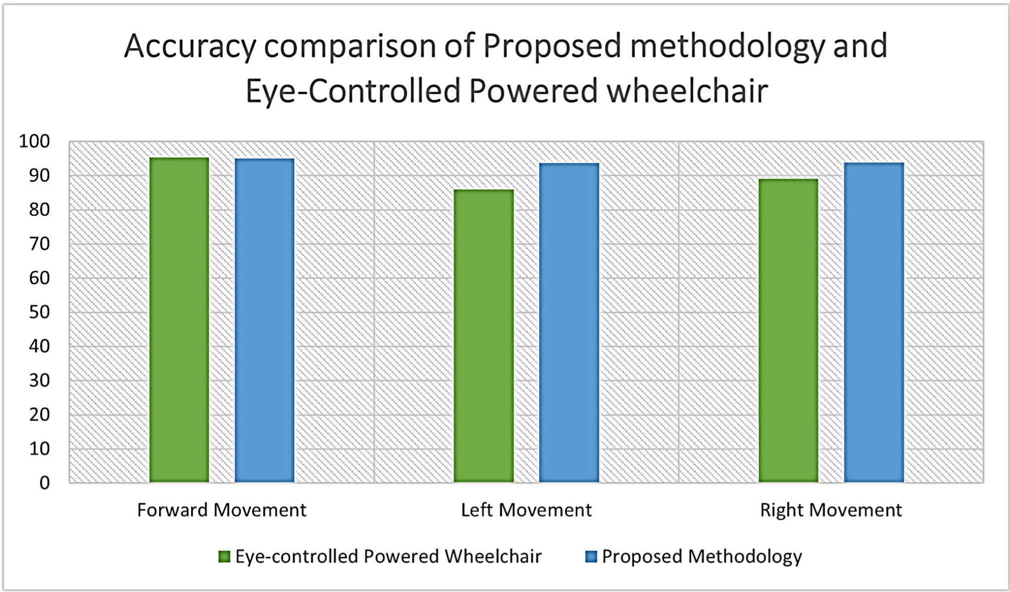

FIGURE 14. Accuracy comparison between our proposed system and eye controlled powered wheelchair.

TABLE 1. Correlating the eye pupils with the direction of movements.

No.	Direction of Movement	Coordinates of Left Pupil	Coordinates of Right Pupil
1	*Looking Right*	(165, 217)	(287, 218)
2	*Looking Centre*	(148, 207)	(265, 207)
3	*Looking Right*	(160, 208)	(283, 211)
4	*Looking Left*	(131, 227)	(217, 227)
5	*Looking Centre*	(139, 202)	(261, 203)
6	*Looking Centre*	(141, 206)	(265, 206)
7	*Looking Left*	(138, 212)	(223, 211)
8	*Looking Right*	(163, 210)	(286, 218)

They can learn from large amounts of training data and capture complex patterns, allowing the smart wheelchair system to handle a wide range of situations. This adaptability is crucial in real-world environments where there can be variations in terrain, obstacles, and user preferences [4], [5], [9], [10].

Coco dataset [11] is being used in our system for object detection. This dataset contains an extensive range of images, each carefully annotated with detailed information about the objects present within them. This level of annotation enables the dataset to support advanced algorithms and models for object detection and recognition. By leveraging this dataset, our system can accurately identify and localize objects within images, allowing for a deeper understanding of the visual content. For our implementation, we specifically work with the Coco 2017 version of the dataset, which encompasses an impressive 123,000 images. This substantial number of images ensures a diverse and representative collection of real-world scenarios, leading to improved generalization and robustness of our object detection system.

B. ACHIEVEMENTS AND CHALLENGES

1) The computational complexities in a vision-based smart wheelchair refer to the challenges and demands placed on the computational systems that process visual data and make decisions to navigate the wheelchair effectively. These complexities arise from the need to analyze real-time visual information and can affect the wheelchair's performance. The proposed system relies on capturing and processing images from cameras in real-time. Analyzing these images quickly and accurately demands significant computational power. Identifying and categorizing objects, such as obstacles, doorways, or people, in the wheelchair's field of view is computationally intensive. The system must continuously analyze and recognize various objects to make informed navigation decisions. Calculating safe paths and avoiding obstacles in real-time is a complex computational task. The system needs to plan routes and adjust them dynamically to prevent collisions. Building and updating a map of the environment, as the

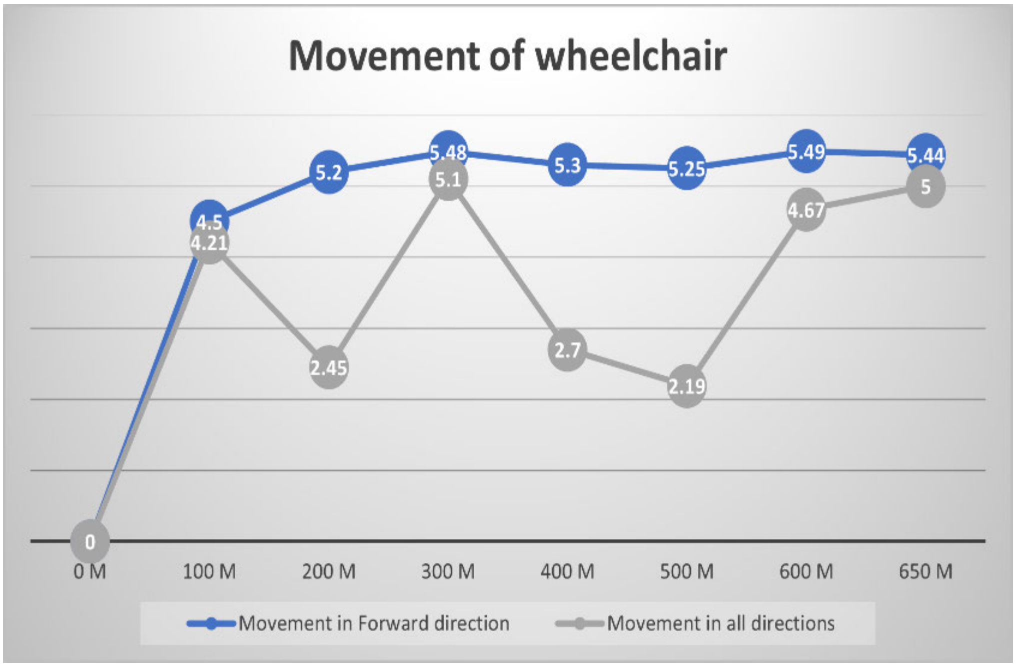

FIGURE 15. Movement of wheelchair in two paths with and without directions.

wheelchair moves, is computationally demanding. This map helps with navigation and requires constant data processing. Smart wheelchairs often use a combination of sensors, including cameras, LIDAR, and ultrasonic sensors. Integrating data from multiple sensor types and processing them efficiently is a computational challenge.

2) The proposed system needs to make quick decisions based on the visual data. It must evaluate various factors, including obstacle proximity, user commands, and navigation goals, in real-time. Computing the optimal path to a destination while considering real-time data and avoiding obstacles involves complex algorithms and continuous updates. Delays in image processing can impact the wheelchair's ability to respond swiftly to changing environments or situations. A slow processing can result in accidents or discomfort for the user. The system needs to store and manage a large amount of visual and mapping data. These computational complexities underscore the need for advanced hardware, software, and algorithms to ensure the smooth and safe operation of vision-based smart wheelchairs. As technology continues to advance, addressing these complexities is crucial to improve the capabilities and reliability of such systems, ultimately benefiting individuals with mobility impairments.
3) The primary advantage of our wheelchair is that it can provide greater independence to users with limited mobility. With traditional joystick-controlled wheelchairs, users must have sufficient upper body strength and dexterity to operate the joystick effectively [3], [12], [13], [14]. However, with a vision-based autonomous wheelchair, users can navigate the environment with minimal physical effort [17], [20], [30].
4) The system can adapt itself based on the user's preferences and abilities. Enhancing the user experience can be achieved by offering customizable controls and options. Users with varying requirements can customize the interface to suit their specific needs. One of the most significant benefits of our wheelchair is that it can help users who have restricted mobility achieve greater levels of independence. It is necessary for users of traditional wheelchairs that are controlled by joysticks to possess adequate upper-body strength and dexterity in order to operate the joystick successfully [3], [12], [13], [14]. Nevertheless, users of a vision-based autonomous wheelchair are able to navigate the world with a minimal amount of physical exertion [35], [39], [48].
5) The methodology of other smart wheelchairs varies depending on the specific technology used. Some smart wheelchairs may use sensors to detect obstacles and provide feedback to the user, while others may incorporate GPS and other navigation technologies to assist with indoor and outdoor navigation. In general, the main goal of smart wheelchairs is to increase the mobility and independence of users with limited mobility. By incorporating advanced technologies such as computer vision and artificial intelligence [31], [57], [58], vision-based autonomous wheelchairs represent a promising approach to achieving this goal.

Vision Based Smart Wheelchair

TARGET / OUTPUT	Looking Right	Looking Centre	Looking Left	SUM
Looking Right	37 33.33%	1 0.90%	0 0.00%	38 97.37% 2.63%
Looking Centre	2 1.80%	23 20.72%	1 0.90%	26 88.46% 11.54%
Looking Left	0 0.00%	4 3.60%	43 38.74%	47 91.49% 8.51%
SUM	39 94.87% 5.13%	28 82.14% 17.86%	44 97.73% 2.27%	103 / 111 92.79% 7.21%

FIGURE 16. **Confusion matrix indicating the efficiency of our system in various dimensions. The overall value of 92.79% is obtained.**

6) Vision-based smart wheelchairs offer many benefits, but they also have several limitations that can impact their effectiveness and usability. Vision-based smart wheelchairs depend on cameras and sensors to "see" their environment. They are limited by poor visibility conditions, such as fog, heavy rain, or low light, which can make navigation challenging or even impossible in certain situations. Navigating through indoor environments, especially in cluttered or complex settings, can be challenging. Vision-based systems may struggle to detect small obstacles, like toys or cables on the floor, which can pose risks to the user. Cameras on wheelchairs have a limited field of view, and they may not see obstacles or objects outside their range. This could result in collisions with objects or people when turning or changing direction. Moreover, users need to learn how to operate and interact with vision-based smart wheelchairs effectively. This learning curve can be steep for some individuals, especially those who are not familiar with technology.
7) The proposed system can be expensive due to the advanced technology that it employs. This cost can limit accessibility for some individuals who may benefit from them. Lastly, as of now, there is no standardization for vision-based smart wheelchairs, which means that different models may have different capabilities and limitations. The users might need to adapt to new systems if they switch to a different wheelchair. In conclusion, while vision-based smart wheelchairs offer promising technological advancements, they are not without limitations. These limitations often revolve around their ability to accurately perceive and respond to the environment, as well as considerations related to privacy, usability, and cost. Addressing these limitations is crucial for making this technology more accessible and effective for individuals with mobility impairments.

VI. CONCLUSION

The idea behind the eye-controlled wheelchair is to enable physically challenged people become more self-sufficient. The purpose of creating an eye-controlled autonomous wheelchair is to show how digital image processing may assist people. In order to reduce the amount of help required by disabled persons, our research proposes a software-hardware co-design solution for wheelchairs. The OpenCV package's digital image processing algorithms included the Haar classifier, Hough transform, Gaussian blur filter, and edge detection. Experiments show that the proposed system can be effectively implemented on a wheelchair prototype, and that the enhanced algorithm reduces execution time by 78% other methodologies like SLAM or Lever based methodologies.

SUGGESTIONS FOR FUTURE WORK

The wheelchair model presented in this work is sufficient and capable in its current state. It is capable of following any curve or cycle. Because points are granted depending on the distance traversed and the whole robot's speed, we must create a wheelchair that is both light and fast. As a result, the circuit was employed with two high-speed motors and high-sensitivity sensors. As mentioned earlier, speed is also affected by body weight and wheel radius. To improve manoeuvrability, we need to create a wheelchair with two motors and two wheels on the back and a free wheel on the front. The power supply has a regulator with a capacity of 12 volts. The wheelchair is equipped with a camera and an ultrasonic sensor that detects obstacles. The Raspberry Pi microcontroller and the BTS 7960 H-bridge motor driver are used to control the direction and speed of the motors. The microcontroller is in charge of controlling the robot. The cost of health care in developing countries like Pakistan is heavily influenced by the land and location of the facility, in addition to the infrastructure and amenities available, as well as the experienced personnel necessary to operate the pricey machinery. This article describes an economic robot with the help of web cam and ultrasonic sensor, a robotics prototype wheelchair for the disabled and the elder people, as per discussion with the relevant physicians with whom we have been in touch.

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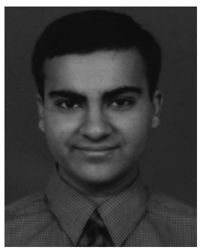

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